ABZ 2027 Case Study: From Reduction to Elimination of Statelessness

Nuno Macedo[1], Alcino Cunha[1], and Patrícia Jerónimo[2]

[1] School of Engineering, University of Minho & INESC TEC
[2] School of Law, University of Minho

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Abstract. The ABZ 2027 case study concerns the modelling and analysis of nationality laws and the protection they provide against statelessness. A shared fictional instance with three States is the required starting point, with justified restrictions permitted. Its laws incorporate safeguards from the 1961 Convention on the Reduction of Statelessness. Participants should model and validate the instance and analyse common scenarios. They are invited to formulate and investigate statelessness elimination first within this instance. Contributions may establish sound abstractions, characterise guarantees and their assumptions, or identify effective legal changes. The general challenge extends the analysis to any nationality laws satisfying the specified safeguards and examines protection in the presence of non-party States. No particular formal method or temporal logic is prescribed.

Keywords: Citizenship · Statelessness · Formal methods · Computational law · Temporal properties · Legal synthesis

1 Motivation and Ambition

A person is stateless when no State considers that person to be its national under the operation of its law. This definition, from Article 1(1) of the 1954 Convention relating to the Status of Stateless Persons, makes statelessness an inherently relational and multi-jurisdictional condition [1]. It may depend on several legal systems, distributed facts, and how authorities apply their laws. A person may have several nationalities, none, or only a route by which nationality could later be acquired.

The 1954 Convention establishes a protection regime for stateless persons; the 1961 Convention addresses prevention and reduction of statelessness[3]. The latter safeguards persons who would otherwise be stateless and are born in a

[3] The 1961 Convention is available at `https://legal.un.org/ilc/texts/instruments/english/conventions/6_1_1961.pdf` and forms an integral part of the case study description. Readers are expected to consult it alongside the case study description.

Contracting State or to one of its nationals, regulates foundlings and births on ships and aircraft, and limits loss and deprivation [2]. It does not prescribe a single nationality law: Article 1 permits automatic acquisition at birth or acquisition upon application subject to specified conditions. This freedom is both legally important and formally challenging.

The case study asks:

> *What protection against statelessness do nationality laws provide, under which assumptions, and when does that protection amount to elimination?*

The word *elimination* is intentionally open. Some readings have easy positive cases; others admit immediate negative examples. The challenge is to make the relevant distinctions precise, characterise when protection holds, and develop abstractions that make sound analysis feasible. A simple answer to one property can coexist with a substantial modelling contribution. No scenario is a counterexample in isolation: its significance depends on the chosen property and assumptions.

Work begins with the shared three-State instance in Section 2. Modelling and validating this instance and analysing common scenarios form the required basis for comparison, subject to explicit and justified restrictions. Section 3 sets out the initial challenge; Section 4 broadens it to families of laws or arbitrary compliant legislation. The wider ambition remains open to diverse methods and creative extensions.

No particular formalism is prescribed. Theorem proving, model checking, rule-based methods, process models, refinement, constraint solving, game- or strategy-based formalisms, and other rigorous approaches are welcome. Choosing a formalism may itself be a contribution where different notions of elimination require different forms of path, knowledge, or agency quantification.

2 Shared Three-State Instance

2.1 Legal context and terminology

A State is a party once it has ratified or acceded and the Convention is in force for it. The 1961 Convention cannot be modelled as a complete citizenship code, leaving States considerable discretion in defining their domestic nationality laws. A principal variation among nationality laws is between *jus soli*, acquisition by birth in a State's territory, and *jus sanguinis*, acquisition by descent from a national. States often combine or qualify these principles. The Convention does not mandate either as a general regime. Instead, it imposes targeted safeguards where a person would otherwise be stateless and restricts specified forms of loss and deprivation. Determining whether a safeguard is triggered may therefore require evaluating several States' ordinary rules. UNHCR guidance observes, for example, that the inquiry for a child includes whether the child acquired a parent's nationality or that of the State of birth [3].

2.2 World and common rules

The following requirements define the world rules for the fixed shared instance:

W1 The world consists of the States of Asteria (A), Borealis (B), and Cyrenia (C), together with persons, discrete time, births, deaths, residence, parenthood, nationality, applications, registrations, and changes of nationality.
W2 All three States are parties to the 1961 Convention, which is in force for them from the initial time.
W3 No other State exists.
W4 Majority is reached at age 18.
W5 A person may hold more than one nationality.
W6 Initial nationality and family relations are input data.
W7 The instance follows *minimal completion*: only the acquisition and loss rules stated below exist, and nationality otherwise persists.
W8 Residence is not exclusive to a single State, does not require a principal or habitual place of residence, and is not subject to a minimum duration.

Under this minimal completion, there is no ordinary naturalisation or nationality effect of marriage or adoption, and no loss except the Borealis retention rule below. These choices fix the background for the common analysis; there are no claims that such legislation is typical, and extensions may vary them explicitly.

Against this baseline, the common rules implement the Convention's safeguards concerning foundlings, births on ships and aircraft, and birth abroad to a national where Article 4 applies. References to Convention provisions identify the rules to be instantiated; submissions should explain how their model represents these rules.

R1 The rules implement the Convention's restrictions on loss and deprivation of nationality under Articles 2–10.
R2 The subsidiary safeguards under Articles 1(4) and 4 operate through automatic acquisition of nationality.
R3 The retained power of deprivation under Article 8(3) is not invoked.
R4 For the purpose of determining Convention obligations, a birth on a ship is treated as a birth in the flag State, and a birth on an aircraft as a birth in the State of registration, as required by Article 3. The vehicle is not treated as additional physical territory for other purposes.

The principal Article 1 acquisition rules are instantiated through the domestic nationality laws specified separately for each State below.

Asteria. Asteria principally follows *jus sanguinis*, qualified by the place of birth:

A1 A child born in Asteria acquires Asterian nationality at birth if at least one parent is Asterian.
A2 A person born in Asteria who would otherwise be stateless may obtain Asterian nationality only by application. This application-based entitlement requires the following conditions permitted by Article 1(2):

A2.1 It must be lodged from age 18 and no later than age 21.
A2.2 The applicant must have habitually resided in Asteria for the five years immediately preceding the application and for ten years in total
A2.3 The applicant must always have been stateless.

Borealis. Borealis principally follows *jus sanguinis*, including for births abroad, supplemented with a restricted *jus soli* safeguard:

B1 A child of a Borealian parent acquires Borealian nationality automatically at birth, wherever born.
B2 Any child born in Borealis who would otherwise be stateless also acquires Borealian nationality automatically at birth under Article 1(1)(a).
B3 A Borealian national born outside Borealis retains that nationality after age 19 only if, at that time, the person resides in Borealis or has registered an intention to retain it. This instantiates the permission in Article 7(5).

Cyrenia. Cyrenia principally follows *jus sanguinis*, but distinguishes births at home from births abroad:

C1 A child born in Cyrenia acquires Cyrenian nationality at birth if at least one parent is Cyrenian.
C2 Under its ordinary descent rule, a child of a Cyrenian parent born abroad acquires nationality only through the immediately available registration entitlement; this does not preclude the automatic acquisition of nationality under any applicable Convention safeguard described above.
C3 A person born in Cyrenia who would otherwise be stateless may obtain Cyrenian nationality on application, following conditions permitted by Article 1(2):
C3.1 The entitlement is not subject to residence or "always stateless" conditions.
C3.2 This application-based entitlement remains open from birth through age 21.

2.3 Scenarios for validation and specification

The scenarios support both model validation and specification design. Applying competing definitions to the same history reveals what each counts as elimination, resolution, or an available route, and whose choices affect that judgement. For every scenario within their scope, submissions should report the represented facts, nationality history, and outcomes for each proposed property. Results should distinguish proofs, bounded checks, concrete counterexamples, and unresolved questions. Alternative interpretations should be identified and justified. The choice of properties and assumptions is part of the initial challenge in Section 3.2.

1. *Automatic prevention.* A child of two stateless parents is born in Borealis. With no conflicting facts, the child should acquire Borealian nationality at birth. This tests the trigger "would otherwise be stateless" and automatic operation of law.

2. *Delayed but successful acquisition.* A child of two stateless parents is born in Asteria, acquires no nationality at birth, resides there continuously, and applies at age 18. The child remains stateless for a period but should then acquire Asterian nationality. This distinguishes immediate prevention from eventual resolution and tests residence and application histories.
3. *Mobility and an unavailable route.* A stateless child of stateless parents is born in Asteria but repeatedly moves among the three States and never accumulates the required five immediately preceding years of residence before Asteria's application window closes. The person has no national parent. This scenario tests whether universal participation alone entails the chosen property and which restrictions on mobility, or which fairness assumptions, would change the result.
4. *Accessible but unused registration.* A child is born in Asteria to a Cyrenian parent and has no nationality at birth. The Cyrenian registration route is available immediately after birth and is entitlement-based: Cyrenia has no discretion to refuse. The parents know about the procedure, face no legal or practical obstacle, and can reasonably be expected to complete it, but do not submit a registration. Under UNHCR guidance, the child remains stateless until registration is completed, but Asteria's Article 1 responsibility is not engaged in these circumstances [3]. The scenario tests whether a model and its chosen protection property distinguish actual statelessness from an accessible but unused route to nationality.
5. *Loss after an opportunity to retain.* A person born in Asteria to a Borealian parent acquires Borealian nationality at birth, remains in Asteria, and does not register an intention to retain the nationality at age 19. The Borealian nationality is lost under its Article 7(5) rule. Although the person satisfies Asteria's residence conditions, an Asterian application cannot satisfy the "always stateless" condition. This tests past nationality, permitted loss, avoidable action, and whether a prevention-oriented Convention entails recovery after statelessness has arisen.
6. *Birth aboard an aircraft.* A child of stateless parents is born aboard an Asterian-registered aircraft while it is in Borealis. For Convention obligations Article 3 treats the birth as occurring in Asteria, so Borealis's automatic territorial safeguard does not apply merely because of the aircraft's physical location. If the child never resides in Asteria, the Asterian application conditions may remain unavailable. This tests jurisdiction, the separation of deemed place of birth from residence, and a less obvious interaction of otherwise compliant rules.
7. *Death before eligibility.* A child of stateless parents is born in Asteria and dies before age 18, having acquired no nationality. This distinguishes acquisition during a lifetime from a conditional future entitlement and from removal by death from the living population.
8. *Continuing births and individual resolution.* Children of stateless parents continue to be born in Asteria arbitrarily far into the future. Each child reaches age 18, satisfies the application conditions, and acquires nationality

at age 18. Analyse whether the proposed property distinguishes resolution for each individual from a time after which no living person is stateless.

Submissions are encouraged to introduce and analyse additional boundary cases or interactions—for example, administrative delay, unavailable evidence, inter-State cooperation, or alternative acquisition rules—and to share promising scenarios before submission through the planned case-study website for discussion, refinement, and reuse.

3 The Initial Challenge

The initial challenge is to model the shared instance and investigate what protection it provides, under explicit assumptions and proposed meanings of elimination.

3.1 Model and validate the shared instance

Let $\mathcal{L}_0$ denote the fixed collection of three fictional nationality laws in Section 2. Every submission must model and analyse a substantive part of this instance, retain the three States, explain how their interacting laws are represented, and report results for each common scenario within its declared scope. Restrictions on provisions, population, histories, or administrative detail are permitted when explicit and justified; omitted scenarios should be identified with reasons. The report should distinguish model-validation outcomes from those that depend on an interpretation or abstraction. Proposed repairs or alternative laws should be analysed as explicit variations of the reference instance.

The reference laws use choices and safeguards supplied by the Convention, so much of their intended conformity follows by construction. Authors are invited to flag inconsistencies, transcription errors, or departures from the intended safeguards as part of validation. A separate proof of Convention compliance is not required for the shared instance.

3.2 Investigate elimination within the fixed instance

Let $\mathcal{E}$ describe environmental assumptions about mobility, births, applications, evidence, choices, and administrative progress, and let $\mathcal{P}$ express a proposed meaning of elimination or protection against statelessness. The initial analysis concerns

$$(\mathcal{L}_0, \mathcal{E}) \models \mathcal{P}.$$

The domestic laws are fixed at this stage. Participants are expected to investigate how their choices of $\mathcal{E}$ and $\mathcal{P}$ affect the result.

One dimension concerns the population and time period covered. A population-wide reading requires a time after which no living person is stateless. An individual reading may require acquisition during each person's lifetime or eventual resolution of every episode of statelessness. Submissions should explain how births and

deaths affect the property: death may remove someone from the population without resolving their statelessness through acquisition of nationality.

A separate dimension concerns the kind of guarantee. Actual acquisition is different from the availability of a legally and practically accessible route: someone who does not use an available registration procedure may remain stateless despite having such a route. For acquisition or retention, participants should distinguish whether the outcome is merely possible, occurs in every admissible evolution, or can be ensured by specified actors, such as the person, parents, authorities, or States. The last interpretation concerns agency or strategic ability, not merely what happens along a single evolution. These choices can be combined with different population and temporal scopes. Submissions should make the relevant quantification over behaviours, choices, and actors explicit.

Some formulations will have easy answers. Unrestricted *jus soli* provides a simple positive comparison, while the shared three-State instance supplies straightforward negative cases for some unconditional lifetime or population-wide claims. Likewise, requiring only that a qualifying application succeed may largely restate the underlying entitlement. Such results remain useful when they clarify the proposed property, expose necessary assumptions, validate a faithful model, or support a reusable abstraction or analysis.

Submissions need not establish a positive elimination result. A contribution may instead formulate one or more plausible stronger notions of elimination and demonstrate how its modelling and verification techniques automatically recover shared scenarios, or generate analogous executions, as counterexamples. Such results should identify the property and assumptions being refuted and explain whether the reported execution is concrete or depends on bounds or abstraction.

3.3 Abstraction and feasibility

The evolving setting also presents an infinite-state challenge. Time and the number of persons may be unbounded, so concrete states need not repeat and finite-state cycle or lasso arguments may not apply directly. Automated analyses should explain their use of bounds, finite abstractions, cutoff results, or complementary proof techniques. In particular, a fixed set of three States does not make an evolving population finite-state. An abstraction that preserves current nationality may still lose the history needed for future eligibility; one that preserves possible acquisition may lose who controls the necessary actions. Authors should explain which conclusions their abstraction preserves and whether reported counterexamples correspond to concrete histories.

4 The General Challenge

After analysing the shared three-State instance, participants are encouraged to investigate how far their results generalise.

4.1 Arbitrary compliant legislation

The wider ambition is to handle an arbitrary finite collection of States and arbitrary domestic nationality laws satisfying a stated formalisation of Convention obligations. Let $\mathcal{C}$ denote those obligations and let $\mathcal{L}$ range over the intended class of laws. The general claim is

$$\forall \mathcal{L}. \quad \mathcal{L} \models \mathcal{C} \quad \Longrightarrow \quad (\mathcal{L}, \mathcal{E}) \models \mathcal{P}.$$

The universal quantification and compliance antecedent enter at this stage. Here compliance defines the class of laws about which a guarantee is sought. The selected obligations $\mathcal{C}$, interpretation of $\mathcal{P}$, and assumption scheme $\mathcal{E}$ should be explicit. If assumptions depend on the legislation, that dependence must be stated. Establishing a property of $\mathcal{L}_0$ alone does not establish this general claim; a failure in $\mathcal{L}_0$ refutes it only if the instance satisfies $\mathcal{C}$ and belongs to the intended class.

Intermediate restrictions are welcome: authors may parameterise the three shared State laws, vary a selected set of acquisition or loss rules, formalise only certain articles, bound numerical parameters, or study other fixed jurisdictions. The report should identify which conclusions concern the shared instance, which hold for a restricted class of nationality laws, and which hold for arbitrary compliant laws. Real legislation may also be studied with explicit sources, dates, and interpretations, alongside the required shared analysis.

Possible questions include whether compliance across States suffices for a chosen global property; which assumptions about applications, residence, and administrative progress are necessary or sufficient; and which guarantees survive the addition of further States or changes in their laws. Contributions may establish monotonicity, dominance, cutoff, or extremal-law results. A rigorous demonstration that a broad class of laws can be represented by one or more extremal cases is an acceptable outcome, even if the remaining analysis is simple. The ordering on laws must be justified for the property: changes in acquisition can affect eligibility for later safeguards.

4.2 Background nationality laws

Beyond the fixed completion in Section 2, authors must state how nationality is determined outside the provisions they model directly. This need not amount to a full comparative nationality code. A useful interface may specify:

- the initial assignment of zero, one, or several nationalities;
- automatic acquisition at birth by descent or territorial connection;
- non-automatic acquisition by application, ordinary naturalisation, registration, declaration, or option, including which decisions are mandatory and which are discretionary;
- retention, renunciation, loss, and deprivation, and whether nationality otherwise persists; and
- the treatment of dual nationality, unknown facts, evidence, and disagreements between authorities.

4.3 Legal reform

Another direction is to identify the weakest changes that establish a chosen property. The easy *jus soli* case in Section 3.2 explains why this question is most informative under explicit constraints—in particular, when States retain *jus sanguinis* as their principal rule and only safeguards, exceptions, or procedures may change. Candidates include automatic acquisition, safeguards for existing stateless persons, limits on decision times, assistance with applications, coordination between States, and restrictions on loss.

Authors should distinguish changes to domestic laws, Convention obligations, and assumptions about the environment in which the model operates, state the admissible reforms and the ordering or cost by which one change is weaker than another, and justify any claim of minimality. Permitted forms of loss must be considered alongside acquisition rules when assessing lasting protection. Necessary and sufficient conditions or a collection of incomparable minimal reforms may be more informative than one sufficient solution.

4.4 Non-party States

An additional extension relaxes universal participation by allowing non-party States in the model. Treaty membership and domestic nationality law should be represented separately: a non-party State may nevertheless provide some or all of the Convention's safeguards.

Participants may investigate under what conditions a chosen protection property survives in such a mixed setting. Relevant questions include which birth, descent, or residence links to participating States are sufficient; whether participating States can compensate for gaps elsewhere; and how much non-participation can be tolerated. Article 4, for example, may protect an otherwise-stateless person born outside a Contracting State when a parent held the nationality of a Contracting State at the person's birth [2]. A global guarantee for persons with no relevant connection to a protective jurisdiction is readily defeated, so the population covered and any additional safeguards should be stated explicitly.

References

1. United Nations: Convention relating to the Status of Stateless Persons. United Nations Treaty Series, vol. 360, p. 117 (1954)
2. United Nations: Convention on the Reduction of Statelessness. United Nations Treaty Series, vol. 989, p. 175 (1961)
3. United Nations High Commissioner for Refugees: Guidelines on Statelessness No. 4: Ensuring Every Child's Right to Acquire a Nationality through Articles 1–4 of the 1961 Convention on the Reduction of Statelessness. HCR/GS/12/04 (2012)